

LAB EXPERIMENT ITA0506-COMPUTER VISION

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EXPNO: 28. Find the Boundary of the Image Using Convolution Kernel

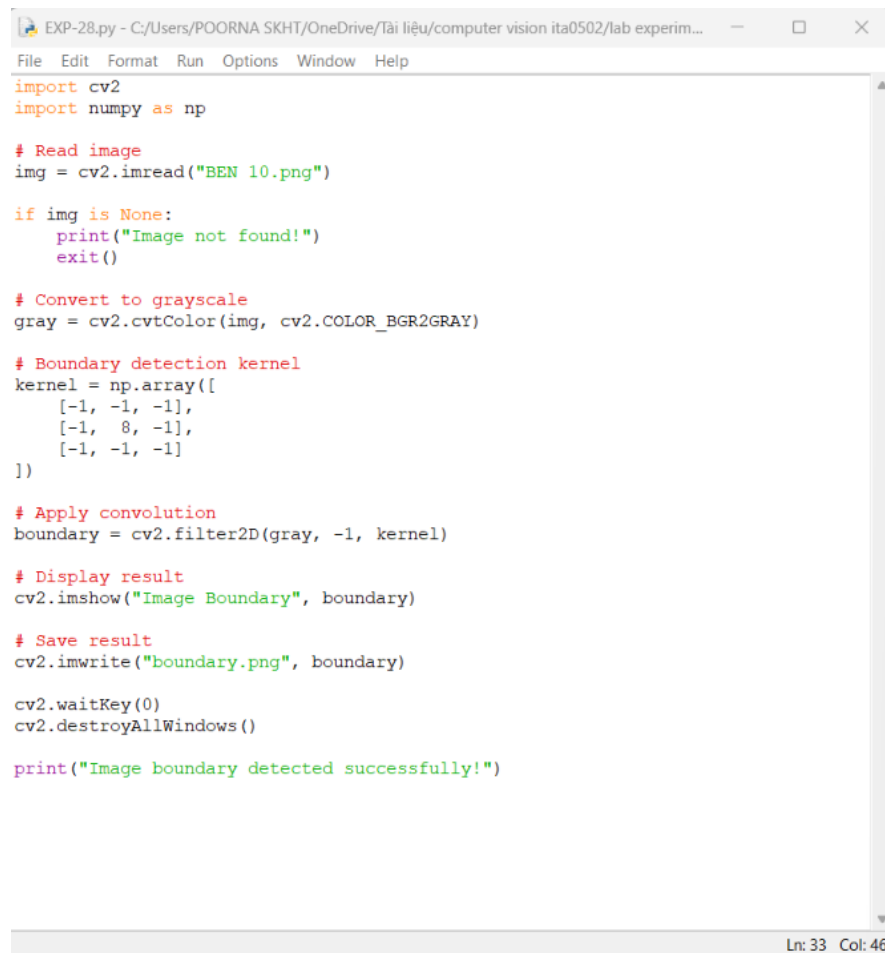
Aim

To find the **boundary/edges of an image using a convolution kernel** in OpenCV.

Algorithm

1. Import OpenCV.
2. Read BEN 10.png.
3. Convert the image to grayscale.
4. Define a convolution kernel.
5. Apply the kernel using cv2.filter2D().
6. Display the detected boundaries.
7. Save the result.

CODE:

A screenshot of a Python IDE window titled 'EXP-28.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...'. The window contains a Python script that reads an image, converts it to grayscale, defines a convolution kernel, applies it, and displays and saves the result. The code is as follows:

```
import cv2
import numpy as np

# Read image
img = cv2.imread("BEN 10.png")

if img is None:
    print("Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Boundary detection kernel
kernel = np.array([
    [-1, -1, -1],
    [-1,  8, -1],
    [-1, -1, -1]
])

# Apply convolution
boundary = cv2.filter2D(gray, -1, kernel)

# Display result
cv2.imshow("Image Boundary", boundary)

# Save result
cv2.imwrite("boundary.png", boundary)

cv2.waitKey(0)
cv2.destroyAllWindows()

print("Image boundary detected successfully!")
```

The status bar at the bottom right of the IDE shows 'Ln: 33 Col: 46'.

OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-26.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-28.py
```



Result

The boundaries of the image are successfully detected using a convolution kernel.